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📍 FAU, Erlangen

Professional Experience

Research Assistant

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

May 2025 – July 2025

Erlangen, Germany

- Engineered machine learning pipelines using scikit-learn to predict brain region volumes from facial features, benchmarking Linear Regression, Ridge Regression, Decision Trees, SVM, and SGD models.
- Constructed a PCA-based statistical shape model for 3D facial feature extraction, enabling efficient dimensionality reduction and feature generation for predictive modeling.
- Conducted comprehensive data analysis, including feature selection, correlation studies, outlier detection, cross-validation, and hyperparameter optimization to enhance model robustness.

Infant Vision Simulation

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

Oct 2024 - Jan 2025

Erlangen, Germany

- Simulated age-dependent visual perception and trained deep learning architectures including VGG and ResNet18 to investigate visual development mechanisms.
- Applied curriculum learning strategies on TinyImageNet and egocentric datasets to emulate progressive infant visual maturation during training.
- Assessed the impact of layer freezing and staged learning approaches on model accuracy and feature representation quality.

Product Manager (Software Development & Automation)

Talents2Germany GmbH

Dec 2023 – Apr 2024

Frankfurt am Main, Germany

- Architected scalable RPA workflows in Python for automated web scraping, data validation, transformation, and database integration.
- Developed backend APIs with Laravel, streamlining communication between frontend applications, backend services, and databases.
- Enhanced an internal LMS platform using Next.js while maintaining technical documentation, deployment procedures, and system workflows.

Bachelor Thesis: LiDAR-Based Localization Evaluation

Technical University of Applied Sciences Würzburg-Schweinfurt

Sep 2022 – Mar 2023

Schweinfurt, Germany

- Investigated and benchmarked LiDAR-based localization algorithms within ROS for autonomous navigation applications.
- Orchestrated data acquisition pipelines leveraging OptiTrack motion capture systems and Unity-based simulation environments.
- Analyzed discrepancies between simulated and real-world sensor data through preprocessing, parameter optimization, and reproducible experimentation.

Software Developer Intern

biz4d - Mentoring Club

Apr 2022 – Sept 2022

Frankfurt am Main, Germany

- Spearheaded a team of five interns in developing a matching algorithm, improving recommendation quality and operational efficiency.
- Containerized SuiteCRM and Laravel applications with Docker and administered Linux servers to ensure reliable deployment and hosting.
- Established Git/GitHub development practices, including branching strategies, code reviews, and collaborative debugging workflows.

Mechatronic System for Part Classification

Kindermann GmbH

Apr 2021 – Aug 2021

Würzburg, Germany

- Delivered an end-to-end mechatronic solution integrating computer vision and Node-RED for real-time industrial part classification.
- Trained ResNet and YOLO-based models, achieving 95% classification accuracy on production datasets.
- Deployed an image acquisition, preprocessing, and inference pipeline on Node-RED, ensuring robust detection performance under varying lighting conditions.

Lab Assistant - Automated Systems and HMI
Technical University of Applied Sciences Würzburg-Schweinfurt

Oct 2021 – Mar 2022
Schweinfurt, Germany

- Supported 45+ students in Node-RED, MQTT, microcontrollers, and embedded systems.
- Built and tested experimental HMI and automation setups, reinforcing theory with practical applications.
- Achieved a 100 % pass rate in written and practical exams through effective mentoring.

Programming Tutor (C++)
Technical University of Applied Sciences Würzburg-Schweinfurt

Apr 2023 – Jul 2023
Schweinfurt, Germany

- Mentored undergraduate students in C++ programming, data structures, algorithms, and operating system fundamentals through tutorials and one-on-one support sessions.
- Designed and delivered practical coding exercises, debugging workshops, and problem-solving sessions to strengthen students' programming and analytical skills.
- Guided students in developing efficient and maintainable C++ solutions, contributing to improved academic performance and deeper understanding of software engineering concepts.

Education

M.Sc. in Autonomy Technology
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Specialization: Human-System Interfaces, Sensor and Perception

Oct 2024 - Expected 2027
Erlangen, Germany

- Focused on autonomous systems, computer vision, machine learning, and intelligent human-machine interaction.
- Completed research-oriented projects in deep learning, vision transformers, diffusion models, robotics perception, and intent detection using Python, PyTorch, and ROS.
- Selected participant in the BEST Course “*The Eye in AI: Understanding Image Recognition*” in Copenhagen, exploring advanced computer vision and image recognition techniques.

B.Eng. in Mechatronics
Technical University of Applied Sciences Würzburg-Schweinfurt (THWS)
Specialization: Automated Systems, Human-Machine Interaction, Automation, and Robotics

Nov 2018 - Aug 2023
Schweinfurt, Germany

- Built strong foundations in robotics, embedded systems, automation, control engineering, and software development.
- Conducted Bachelor's thesis evaluating LiDAR-based localization algorithms using ROS, Unity simulation, and OptiTrack systems.
- Completed industrial projects involving computer vision, Node-RED, machine learning, IoT, and mechatronic system design.

Skills

Programming & Scripting: Python, C++, C, JavaScript, MATLAB, Bash.

Machine Learning & Computer Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV, CNNs, ResNet, VGG, YOLO, PCA, Regression, Classification, Model Evaluation, 3D data handling.

Robotics & Autonomous Systems: ROS, LiDAR-based Localization, Sensor Data Processing, 3D Reconstruction, ICP, OptiTrack, Unity Simulation

Web, Backend & Automation: Next.js, Laravel, API development, Web Scraping, Database Integration, SQL, ChromaDB

DevOps & Tools: Git, GitHub, Docker, Linux, CMake, CI/CD basics, Server Configuration, Debugging, Documentation, Agile Workflows

Embedded & Industrial Systems: Node-RED, MQTT, Microcontrollers, Embedded Systems, IoT, Image Acquisition Pipelines

Soft Skills: Team Leadership, Technical documentation, Project Management, Problem Solving, Analytical Thinking, Research-Oriented Working Style.

Languages

English: Fluent (C2)

Konkani: Mother tongue

German: Intermediate (B1)